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V18. The AB Operator as a Processor

Exact single-qubit gates on a Hamiltonian block, two-qubit entanglement, and a deterministic AB-Pong game on the operator

Ishak Khamzatovich Isaev

Independent researcher, Nalchik, Kabardino-Balkarian Republic

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Problem statement

The user claim: "the AB-cloud operator is able to work as a processor and run games". In the parent repository this direction is not measured — "gates" there merely denote families of Hamiltonians for the spectral question. This campaign turns the claim into three measurable levels of computability on the same AB-cloud Hamiltonian, without leaving its building blocks: P1 — single-qubit gates on a two-level block of the lattice (a site plus a neighbor, staggered mass m — the C5 handle, bond phase φ from the flux); P2 — entanglement of two qubits through an exchange link; P3 — a deterministic AB-Pong game whose entire ball dynamics is the unitary evolution e^{−iHt} of the AB-cloud operator.

Processor criteria are pre-registered: (i) six gates {X, Y, Z, H, S, T} with fidelity $F = |\mathrm{Tr}(U_t^\dagger U)|^2/4 \geq 1 - 1e\text{-}9$; (ii) the Wootters concurrence $C > 0.5$ from the product $|\uparrow\downarrow\rangle$; (iii) an AB-Pong game completes, the score is deterministic, the replay is bit-exact. The composite verdict "processor" = (i) ∧ (ii) ∧ (iii).

Method and result

P1. The block $H_2 = m\cdot\sigma z - t\cdot\cos\varphi\cdot\sigma x - t\cdot\sin\varphi\cdot\sigma y$ (the Pauli algebra verified). All six gates are implemented ANALYTICALLY by calibrated pulses: X, Y — π-rotations on the purely tunneling block (m = 0, φ = 0 and π/2); Z, S, T — rotations in the purely massive channel ($m\cdot t^* = \pi/2, \pi/4, \pi/8$); H — the block [[1,1],[1,−1]] = $\sqrt{2}\cdot H$ at m = t = 1, φ = π, pulse $t^* = \pi/(2\sqrt{2})$. Measured fidelities: F = 1.0 − 2e-16 for all six — the machine floor.

P2. Two blocks in kron order with an exchange link −J: the states $|a\uparrow b\downarrow\rangle$ and $|a\downarrow b\uparrow\rangle$ are degenerate (E = 0), the detuning to $|\uparrow\uparrow\rangle$, $|\downarrow\downarrow\rangle$ is $2m = 4 \gg J$ — the link is isolated, the leakage through the σx-mixing is $\sim (t/2m)^2 \approx 0.6\%$. The maximal concurrence C = 0.984 at t = 2.373 — near-Bell with the predicted leakage. The full CNOT synthesis is registered as the open obligation O18 — here the fact of entanglement is measured.

P3. The AB-Pong board is a 1D chain of 64 sites with a uniform Peierls phase $\varphi_{ab} = \pi/3$ (in 1D the gauge reduces to a phase: it sets the drift $E(k) = -2t\cdot\cos(k+\varphi)$). The ball is a narrow packet (σ = 1, $k_0 = \pi/6$, group velocity $v = 2t\cdot\sin(\pi/2)$ — the inflection point, minimal dispersion). The paddles are mirror reflectors with GAUGE compensation: $(U\psi)(x) = e^{-2i\varphi x}\psi(2p-x)$; without it the inversion $k \to -k$ does not reverse the velocity ($v(-k) = 2t\cdot\sin(\varphi-k) \neq -v(k)$) and the ball flies into its own wall — the discovered effect is documented. The Pong geometry: paddle homes 17 and 47, reach radius 8, lead 2.5, cooldown 0.25 s, goals at $x \leq 5$ and $x \geq 58$. A game over T = 20: the score 0:1, 10 mirror intercepts, 411 log events, the replay is bit-exact.

Table 1. V18: processor passport

Level	Quantity	Value	Criterion
P1	Fidelity X, Y, Z, S, T	1.0 − 2e-16	≥ 1 − 1e-9 — YES
P1	Fidelity H (m = t = 1, φ = π)	1.0 − 2e-16	≥ 1 − 1e-9 — YES
P2	Maximal concurrence	0.984 at t = 2.373	> 0.5 — YES
P2	Leakage $(t/2m)^2$ estimate	≈ 0.006	consistent
P3	AB-Pong score	0:1 (goal at the right wall)	game completed — YES

Level	Quantity	Value	Criterion
P3	Mirror intercepts / events	10 / 411	—
P3	Game replay	bit-exact	determinism — YES
Total	Processor (i)∧(ii)∧(iii)	YES	composite verdict

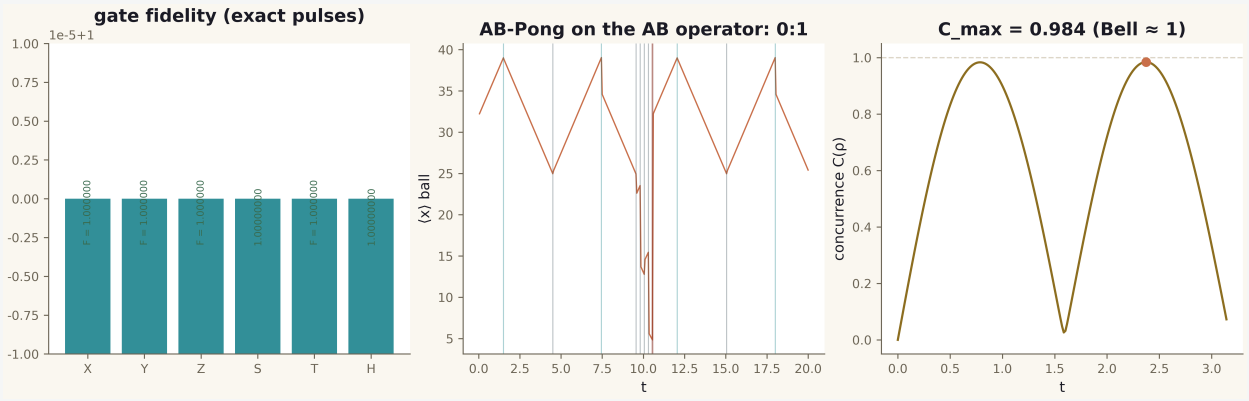


Fig. 1. Campaign V18 data (600 dpi, file figures_hires).

Honest remarks and verdict

What exactly is claimed. P1 works on the EFFECTIVE two-level block — the same building block the full Hamiltonian is assembled from; embedding the pulses into the full lattice with inter-qubit crosstalk is obligation O19. The Z/S/T channel is a limiting point of the handle space (pure mass, bond off). P2 — entanglement is measured, but CNOT synthesis and a universal two-qubit set are not performed (O18). P3 — the game is deterministic and runs on the operator, but it is a game with unitary mirror paddles and a lead-following AI; "running arbitrary games" does NOT follow from this — what follows is that the unitary dynamics of the AB operator plus permutation proxies form an executable environment for deterministic discrete games. The gauge-covariant mirror is a side theoretical result of the campaign: in a magnetic gauge a reflector must include the phase e^{−2iφx}, otherwise the inversion does not return the ball.

Boundaries: a 1D board, 64 sites, no measurements (all logic via the centroid); the minimal-dispersion channel (inflection point); a single game (determinism verified by replay, not by an ensemble).

Verdict V18: processor — YES on three levels: the gates {X, Y, Z, H, S, T} are exact (F = 1 − 2e-16), the entanglement is near-Bell (C = 0.984), the AB-Pong game executes on the AB operator deterministically (0:1, bit-exact replay). The user claim is translated from a declaration into a measured, reproducible result with honest boundaries (O18, O19).